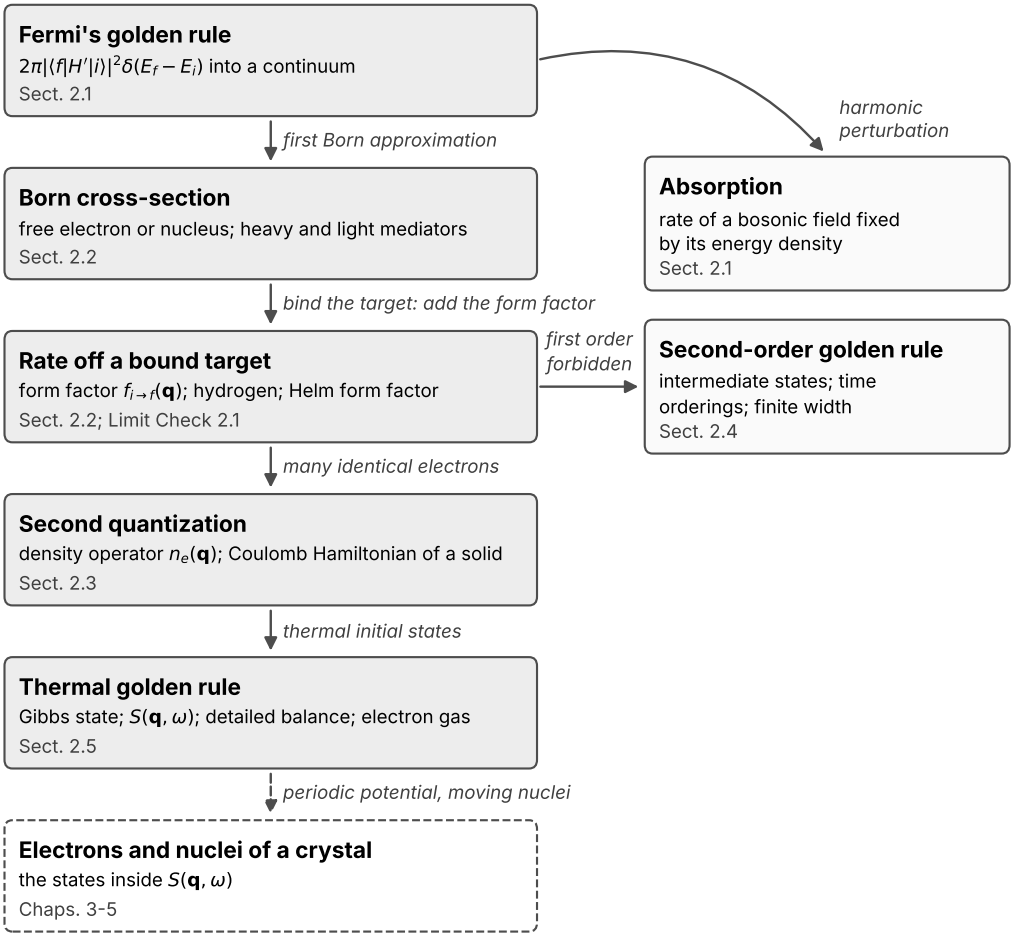
Fermi's golden rule
$2\pi|\langle f|H'|i\rangle|^2\delta(E_f - E_i)$ into a continuum
Sect. 2.1
harmonic perturbation
first Born approximation
Absorption
rate of a bosonic field fixed by its energy density
Sect. 2.1
Born cross-section
free electron or nucleus; heavy and light mediators
Sect. 2.2
bind the target: add the form factor
Rate off a bound target
form factor $f_{i\to f}(\mathbf{q})$; hydrogen; Helm form factor
Sect. 2.2; Limit Check 2.1
first order forbidden
Second-order golden rule
intermediate states; time orderings; finite width
Sect. 2.4
many identical electrons
Second quantization
density operator $n_e(\mathbf{q})$; Coulomb Hamiltonian of a solid
Sect. 2.3
thermal initial states
Thermal golden rule
Gibbs state; $S(\mathbf{q}, \omega)$; detailed balance; electron gas
Sect. 2.5
periodic potential, moving nuclei
Electrons and nuclei of a crystal
the states inside $S(\mathbf{q}, \omega)$
Chaps. 3-5